Predictive Modelling in Excel

Guest Nights and Local Government Tax Revenue (Canada, 1980-2024)
Business Intelligence (2030AFE) - Assignment 2, Question 3 | Tien Dat Ly

Business Context

For many local governments in Canada, tourism is a vital source of revenue, and one way that revenue is generated is by taxing hotel stays. A tourism hotel operator wants to understand the relationship between **guest nights** and **local government tax revenue** so that future tax revenue can be predicted. Annual data for both variables from 1980 to 2024 (45 observations) was analysed in Excel using correlation and simple linear regression.

1. Exploring the Relationship

	Guest nights ('000)	*Tax revenue ($m)*
Guest nights ('000)	1	
Tax revenue ($m)	0.898781924	1

Figure 1: Correlation matrix (Excel)

- **Strength:** The correlation coefficient of 0.8988 is very close to +1, indicating a very strong linear relationship.
- **Direction:** The positive sign means that as guest nights increase, tax revenue also tends to increase.

The result suggests that the link between tourist activity and local government revenue is direct and highly consistent. A large proportion of the year-to-year variation in tax revenue is explained by variation in guest nights.

2. Visualisation

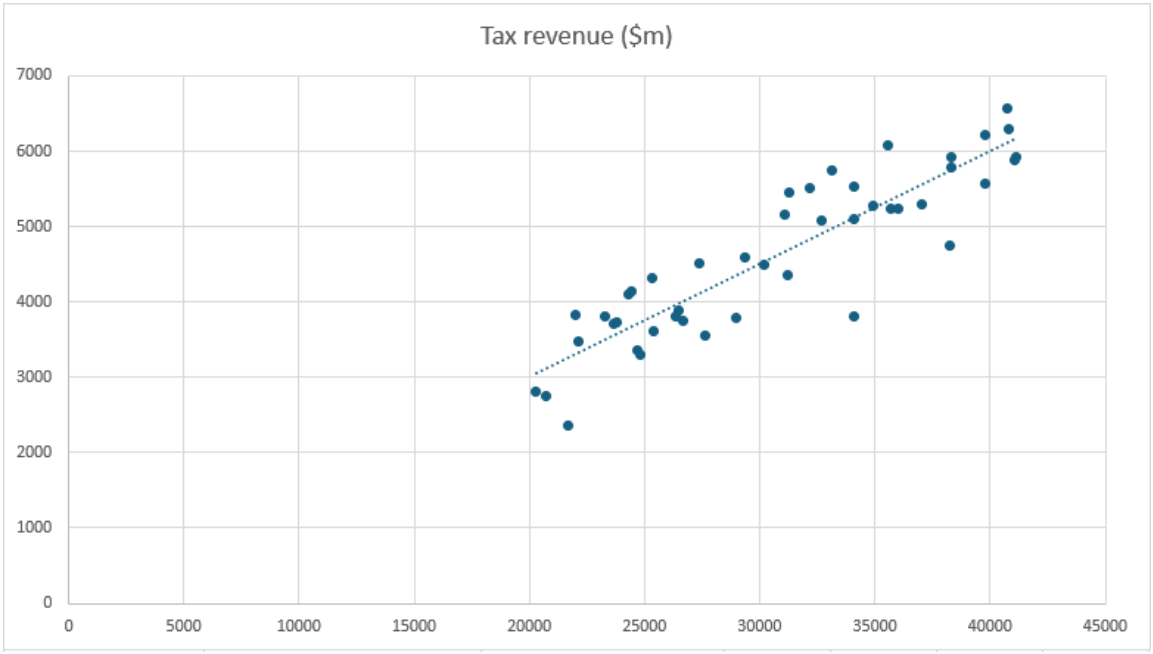

Figure 2: Scatter plot of guest nights ('000) vs. tax revenue ($m) with linear trendline

- **Strength:** The data points cluster tightly around the trendline, reinforcing the very strong linear relationship. Guest nights is an excellent indicator for predicting tax revenue.

- **Direction:** The points rise from the lower-left to the upper-right of the plot, showing that higher guest nights are associated with higher hotel tax revenue.

3. Predictive Model

Model development

Simple linear regression was used because there is one independent variable (guest nights) and one dependent variable (tax revenue).

$$\textbf{Tax Revenue} = \textbf{Beta}_0 + \textbf{Beta}_1 \times \textbf{Guest Nights} + \textbf{Error}$$

- **Intercept ($Beta_0$) = 33.5545:** the estimated tax revenue when guest nights are zero.
- **Guest nights ($Beta_1$) = 0.1492:** for every increase of 1,000 guest nights, expected tax revenue rises by about $0.1492m.

Estimated regression equation: y = 33.5545 + 0.1492x

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.898781924
R Square	0.807808947
Adjusted R Square	0.803339388
Standard Error	472.9393207
Observations	45

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	40425443.73	40425443.73	180.7357015	5.33362E-17
Residual	43	9617878.847	223671.6011		
Total	44	50043322.58			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	33.55451459	347.682471	0.096509078	0.923564674	-667.6140125	734.7230417	-667.614012	734.7230417
Guest nights ('000)	0.149208527	0.011098689	13.44379788	5.33362E-17	0.126825888	0.171591165	0.126825888	0.171591165

Figure 3: Excel regression summary output

Key outputs

Statistic	Value	Interpretation
Multiple R	0.8988	Strong, positive linear relationship
R Square	0.8078	80.78% of the variation in tax revenue is explained by guest nights (very good fit)
Adjusted R Square	0.8033	Still very high after adjusting for the number of predictors
Standard Error	472.94	Average distance of observed points from the regression line
Observations	45	Annual data, 1980-2024
Significance F / p-value	5.33E-17	Far below 0.05, so the model is statistically significant; reject H0 that the slope is zero

Model assessment and interpretation

Higher numbers of guest nights are **strongly and significantly** linked with higher tax revenues.

- The guest nights coefficient is positive: as guest nights increase, predicted tax revenue also increases.
- For every 1,000 additional guest nights, local government can expect about $0.1492m more in tax revenue.

- The high R^2 shows the model explains most of the variation in tax revenue, which supports confidence in the predictive relationship.
- The extremely small p-value provides strong evidence that the relationship is not due to random chance.

Application

The model can forecast expected tax revenue from projected guest nights, helping local government budget and plan public services. If there is an initiative to increase tourism, the model can estimate the expected return in tax revenue.

Limitations

- The model is simple and ignores other factors that could affect tax revenue, such as tax rate changes and economic conditions.
- It is based on 45 annual observations (1980-2024). Predicting for guest night levels well outside the observed range is risky, because the relationship may not hold there.

Tools: Microsoft Excel (correlation, scatter plot with trendline, regression summary output). Technique: correlation analysis and simple linear regression.